4.3 CONTROLLED FULL-WAVE RECTIFIERS

A versatile method of controlling the output of a full-wave rectifier is to substitute controlled switches such as thyristors (SCRs) for the diodes. Output is controlled by adjusting the delay angle of each SCR, resulting in an output voltage that is adjustable over a limited range.

Controlled full-wave rectifiers are shown in Fig. 4-10. For the bridge rectifier, SCRs S_1 and S_2 will become forward-biased when the source becomes positive but will not conduct until gate signals are applied. Similarly, S_3 and S_4 will become forward-biased when the source becomes negative but will not conduct until they receive gate signals. For the center-tapped transformer rectifier, S_1 is forward-biased when v_s is positive, and S_2 is forward-biased when v_s is negative, but each will not conduct until it receives a gate signal.

The delay angle α is the angle interval between the forward biasing of the SCR and the gate signal application. If the delay angle is zero, the rectifiers behave exactly as uncontrolled rectifiers with diodes. The discussion that follows generally applies to both bridge and center-tapped rectifiers.

Resistive Load

The output voltage waveform for a controlled full-wave rectifier with a resistive load is shown in Fig. 4-10*c*. The average component of this waveform is determined from

$$V_o = \frac{1}{\pi}\int_{\alpha}^{\pi} V_m \sin(\omega t)\, d(\omega t) = \frac{V_m}{\pi}(1 + \cos\alpha) \tag{4-23}$$

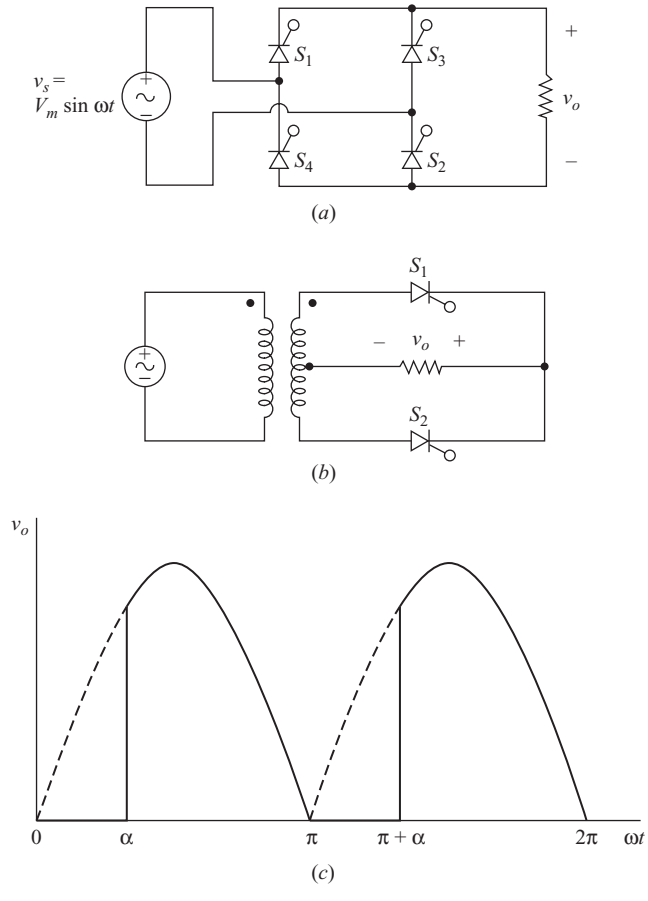


Figure 4-10 (*a*) Controlled full-wave bridge rectifier; (*b*) Controlled full-wave center-tapped transformer rectifier; (*c*) Output for a resistive load.

Average output current is then

$$I_o = \frac{V_o}{R} = \frac{V_m}{\pi R}(1 + \cos \alpha) \tag{4-24}$$

The power delivered to the load is a function of the input voltage, the delay angle, and the load components; $P = I_{\text{rms}}^2 R$ is used to determine the power in a resistive load, where

$$\begin{aligned} I_{\text{rms}} &= \sqrt{\frac{1}{\pi}\int_{\alpha}^{\pi}\left(\frac{V_m}{R}\sin \omega t\right)^2 d(\omega t)} \\ &= \frac{V_m}{R}\sqrt{\frac{1}{2} - \frac{\alpha}{2\pi} + \frac{\sin(2\alpha)}{4\pi}} \end{aligned} \tag{4-25}$$

The rms current in the source is the same as the rms current in the load.

EXAMPLE 4-6

Controlled Full-Wave Rectifier with Resistive Load

The full-wave controlled bridge rectifier of Fig. 4-10*a* has an ac input of 120 V rms at 60 Hz and a 20-Ω load resistor. The delay angle is 40°. Determine the average current in the load, the power absorbed by the load, and the source voltamperes.

■ Solution

The average output voltage is determined from Eq. (4-23).

$$V_o = \frac{V_m}{\pi}\left(1 + \cos\alpha\right) = \frac{\sqrt{2}\,(120)}{\pi}\left(1 + \cos 40°\right) = 95.4 \text{ V}$$

Average load current is

$$I_o = \frac{V_o}{R} = \frac{95.4}{20} = 4.77 \text{ A}$$

Power absorbed by the load is determined from the rms current from Eq. (4-24), remembering to use α in radians.

$$I_{\text{rms}} = \frac{\sqrt{2}(120)}{20}\sqrt{\frac{1}{2} - \frac{0.698}{2\pi} + \frac{\sin[2(0.698)]}{4\pi}} = 5.80 \text{ A}$$

$$P = I_{\text{rms}}^2 R = (5.80)^2\,(20) = 673 \text{ W}$$

The rms current in the source is also 5.80 A, and the apparent power of the source is

$$S = V_{\text{rms}} I_{\text{rms}} = (120)(5.80) = 696 \text{ VA}$$

Power factor is

$$\text{pf} = \frac{P}{S} = \frac{672}{696} = 0.967$$

RL Load, Discontinuous Current

Load current for a controlled full-wave rectifier with an *RL* load (Fig. 4-11*a*) can be either continuous or discontinuous, and a separate analysis is required for each. Starting the analysis at $\omega t = 0$ with zero load current, SCRs S_1 and S_2 in the bridge rectifier will be forward-biased and S_3 and S_4 will be reverse-biased as the source voltage becomes positive. Gate signals are applied to S_1 and S_2 at $\omega t = \alpha$, turning S_1 and S_2 on. With S_1 and S_2 on, the load voltage is equal to the source voltage. For this condition, the circuit is identical to that of the controlled half-wave rectifier of Chap. 3, having a current function

$$i_o(\omega t) = \frac{V_m}{Z}\left[\sin(\omega t - \theta) - \sin(\alpha - \theta)\, e^{-(\omega t - \alpha)/\omega\tau}\right] \qquad \text{for } \alpha \le \omega t \le \beta$$

where (4-26)

$$Z = \sqrt{R^2 + (\omega L)^2} \qquad \theta = \tan^{-1}\left(\frac{\omega L}{R}\right) \qquad \text{and} \qquad \tau = \frac{L}{R}$$

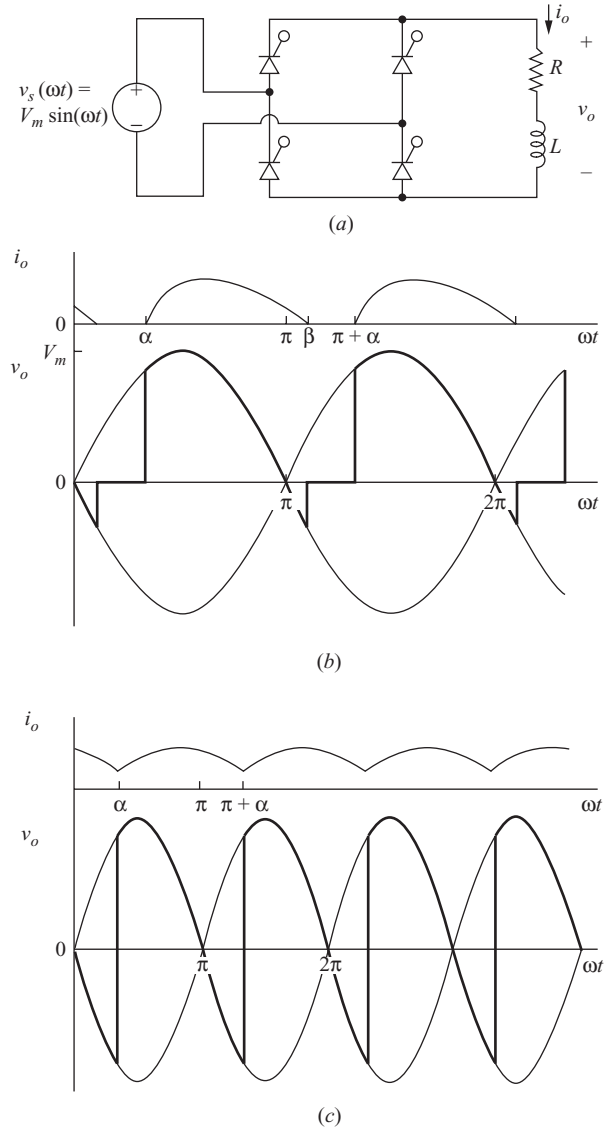


Figure 4-11 (*a*) Controlled rectifier with *RL* load; (*b*) Discontinuous current; (*c*) Continuous current.

The above current function becomes zero at $\omega t = \beta$. If $\beta < \pi + \alpha$, the current remains at zero until $\omega t = \pi + \alpha$ when gate signals are applied to S_3 and S_4 which are then forward-biased and begin to conduct. This mode of operation is called *discontinuous current*, which is illustrated in Fig. 4-11*b*.

$$\beta < \alpha + \pi \rightarrow \text{discontinuous current} \tag{4-27}$$

Analysis of the controlled full-wave rectifier operating in the discontinuous-current mode is identical to that of the controlled half-wave rectifier except that the period for the output current is π rather than 2π rad.

EXAMPLE 4-7

Controlled Full-Wave Rectifier, Discontinuous Current

A controlled full-wave bridge rectifier of Fig. 4-11*a* has a source of 120 V rms at 60 Hz, $R = 10\ \Omega$, $L = 20$ mH, and $\alpha = 60°$. Determine (*a*) an expression for load current, (*b*) the average load current, and (*c*) the power absorbed by the load.

■ Solution

From the parameters given,

$$V_m = \frac{120}{\sqrt{2}} = 169.7 \text{ V}$$

$$Z = \sqrt{R^2 + (\omega L)^2} = \sqrt{10^2 + [(377)(0.02)]^2} = 12.5\ \Omega$$

$$\theta = \tan^{-1}\left(\frac{\omega L}{R}\right) = \tan^{-1}\left[\frac{(377)(0.02)}{10}\right] = 0.646 \text{ rad}$$

$$\omega\tau = \frac{\omega L}{R} = \frac{(377)(0.02)}{10} = 0.754 \text{ rad}$$

$$\alpha = 60° = 1.047 \text{ rad}$$

(*a*) Substituting into Eq. (4-26),

$$i_o(\omega t) = 13.6 \sin(\omega t - 0.646) - 21.2e^{-\omega t/0.754} \text{ A} \qquad \text{for } \alpha \le \omega t \le \beta$$

Solving $i_o(\beta) = 0$ numerically for β, $\beta = 3.78$ rad (216°). Since $\pi + \alpha = 4.19 > \beta$, the current is discontinuous, and the above expression for current is valid.

(*b*) Average load current is determined from the numerical integration of

$$I_o = \frac{1}{\pi}\int_{\alpha}^{\beta} i_o(\omega t)\, d(\omega t) = 7.05 \text{ A}$$

(*c*) Power absorbed by the load occurs in the resistor and is computed from $I_{\text{rms}}^2 R$, where

$$I_{\text{rms}} = \sqrt{\frac{1}{\pi}\int_{\alpha}^{\beta} i_o(\omega t)\, d(\omega t)} = 8.35 \text{ A}$$

$$P = (8.35)^2(10) = 697 \text{ W}$$

10.2 SINGLE-PHASE FULL CONVERTERS

The circuit arrangement of a single-phase full converter is shown in Figure 10.1a with a highly inductive load so that the load current is continuous and ripple free [10]. During the positive half-cycle, thyristors T_1 and T_2 are forward biased; when these two

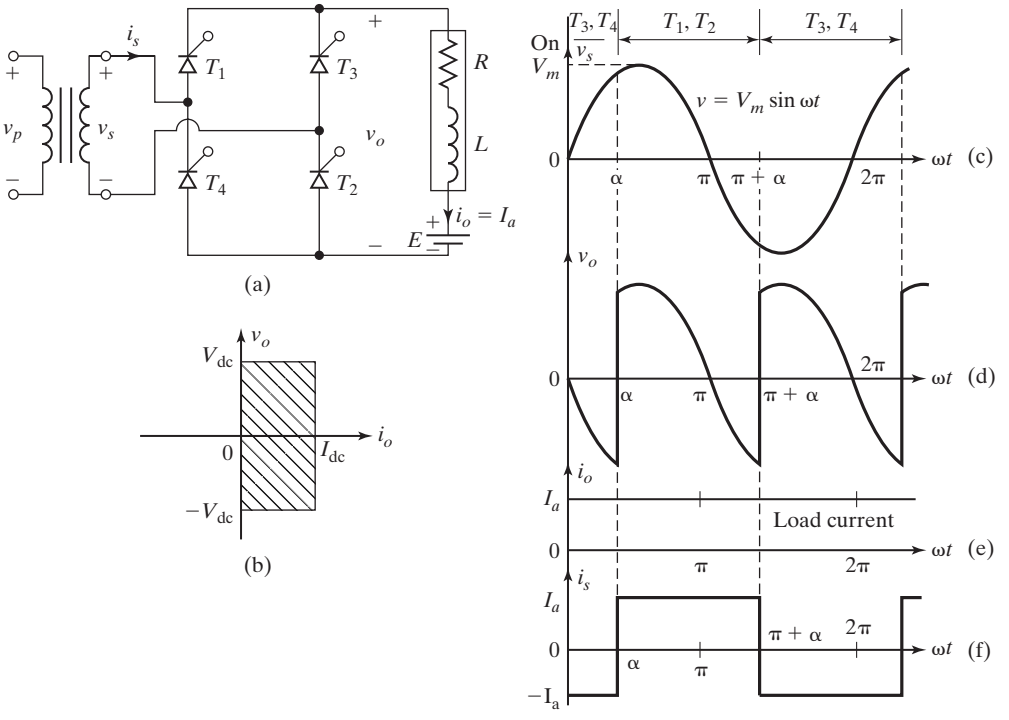


FIGURE 10.1

Single-phase full converter. (a) Circuit, (b) Quadrant, (c) Input supply voltage, (d) Output voltage, (e) Constant load current, and (f) Input supply current.

thyristors are turned on simultaneously at $\omega t = \alpha$, the load is connected to the input supply through T_1 and T_2. Due to the inductive load, thyristors T_1 and T_2 continue to conduct beyond $\omega t = \pi$, even though the input voltage is already negative. During the negative half-cycle of the input voltage, thyristors T_3 and T_4 are forward biased; the turning on of thyristors T_3 and T_4 applies the supply voltage across thyristors T_1 and T_2 as reverse blocking voltage. T_1 and T_2 are turned off due to *line* or *natural commutation* and the load current is transferred from T_1 and T_2 to T_3 and T_4. Figure 10.1b shows the regions of converter operation and Figures 10.1c–f show the waveforms for input voltage, output voltage, and input and output currents.

During the period from α to π, the input voltage v_s and input current i_s are positive, and the power flows from the supply to the load. The converter is said to be operated in *rectification* mode. During the period from π to $\pi + \alpha$, the input voltage v_s is negative and the input current i_s is positive, and reverse power flows from the load to the supply. The converter is said to be operated in *inversion mode*. This converter is extensively used in industrial applications up to 15 kW [1]. Depending on the value of α, the average output voltage could be either positive or negative and it provides two-quadrant operation.

The average output voltage can be found from

$$V_{\text{dc}} = \frac{2}{2\pi}\int_{\alpha}^{\pi+\alpha} V_m \sin \omega t \, d(\omega t) = \frac{2V_m}{2\pi}[-\cos \omega t]_{\alpha}^{\pi+\alpha}$$
$$= \frac{2V_m}{\pi}\cos \alpha \tag{10.1}$$

and V_{dc} can be varied from $2V_m/\pi$ to $-2V_m/\pi$ by varying α from 0 to π. The maximum average output voltage is $V_{dm} = 2V_m/\pi$ and the normalized average output voltage is

$$V_n = \frac{V_{\text{dc}}}{V_{dm}} = \cos \alpha \tag{10.2}$$

The rms value of the output voltage is given by

$$V_{\text{rms}} = \left[\frac{2}{2\pi}\int_{\alpha}^{\pi+\alpha} V_m^2 \sin^2 \omega t \, d(\omega t)\right]^{1/2} = \left[\frac{V_m^2}{2\pi}\int_{\alpha}^{\pi+\alpha} (1 - \cos 2\omega t) \, d(\omega t)\right]^{1/2}$$
$$= \frac{V_m}{\sqrt{2}} = V_s \tag{10.3}$$

With a purely resistive load, thyristors T_1 and T_2 can conduct from α to π, and thyristors T_3 and T_4 can conduct from $\alpha + \pi$ to 2π.

Example 10.1 Finding the Input Power Factor of a Single-Phase Full Converter

The full converter in Figure 10.1a is connected to a 120-V, 60-Hz supply. The load current I_a is continuous and its ripple content is negligible. The turns ratio of the transformer is unity. (a) Express the input current in a Fourier series; determine the HF of the input current, DF, and input PF. (b) If the delay angle is $\alpha = \pi/3$, calculate V_{dc}, V_n, V_{rms}, HF, DF, and PF.

Solution

a. The waveform for input current is shown in Figure 10.1c and the instantaneous input current can be expressed in a Fourier series as

$$i_s(t) = a_0 + \sum_{n=1,2,\ldots}^{\infty} (a_n \cos n\omega t + b_n \sin n\omega t)$$

where

$$a_0 = \frac{1}{2\pi}\int_{\alpha}^{2\pi+\alpha} i_s(t) \, d(\omega t) = \frac{1}{2\pi}\left[\int_{\alpha}^{\pi+\alpha} I_a \, d(\omega t) - \int_{\pi+\alpha}^{2\pi+\alpha} I_a \, d(\omega t)\right] = 0$$

$$a_n = \frac{1}{\pi}\int_{\alpha}^{2\pi+\alpha} i_s(t) \cos n\omega t \, d(\omega t)$$
$$= \frac{1}{\pi}\left[\int_{\alpha}^{\pi+\alpha} I_a \cos n\omega t \, d(\omega t) - \int_{\pi+\alpha}^{2\pi+\alpha} I_a \cos n\omega t \, d(\omega t)\right]$$

$$= -\frac{4I_a}{n\pi}\sin n\alpha \quad \text{for } n = 1, 3, 5, \ldots$$
$$= 0 \quad \text{for } n = 2, 4, \ldots$$
$$b_n = \frac{1}{\pi}\int_{\alpha}^{2\pi+\alpha} i(t)\,\sin n\omega t\, d(\omega t)$$
$$= \frac{1}{\pi}\left[\int_{\alpha}^{\pi+\alpha} I_a \sin n\omega t\, d(\omega t) - \int_{\pi+\alpha}^{2\pi+\alpha} I_a \sin n\omega t\, d(\omega t)\right]$$
$$= \frac{4I_a}{n\pi}\cos n\alpha \qquad \text{for } n = 1, 3, 5, \ldots$$
$$= 0 \qquad \text{for } n = 2, 4, \ldots$$

Because $a_0 = 0$, the input current can be written as

$$i_s(t) = \sum_{n=1,3,5\ldots}^{\infty} \sqrt{2}\, I_n \sin(n\omega t + \phi_n)$$

where

$$\phi_n = \tan^{-1}\frac{a_n}{b_n} = -n\alpha \tag{10.4}$$

and ϕ_n is the displacement angle of the nth harmonic current. The rms value of the nth harmonic input current is

$$I_{sn} = \frac{1}{\sqrt{2}}(a_n^2 + b_n^2)^{1/2} = \frac{4I_a}{\sqrt{2}\,n\pi} = \frac{2\sqrt{2}\,I_a}{n\pi} \tag{10.5}$$

and the rms value of the fundamental current is

$$I_{s1} = \frac{2\sqrt{2}\,I_a}{\pi}$$

The rms value of the input current can be calculated from Eq. (10.5) as

$$I_s = \left(\sum_{n=1,3,5,\ldots}^{\infty} I_{sn}^2\right)^{1/2}$$

I_s can also be determined directly from

$$I_s = \left[\frac{2}{2\pi}\int_{\alpha}^{\pi+\alpha} I_a^2\, d(\omega t)\right]^{1/2} = I_a$$

From Eq. (3.22) the HF is found as

$$\text{HF} = \left[\left(\frac{I_s}{I_{s1}}\right)^2 - 1\right]^{1/2} = 0.483 \quad \text{or} \quad 48.3\%$$

From Eqs. (3.21) and (10.4), the DF is

$$\text{DF} = \cos\phi_1 = \cos(-\alpha) \tag{10.6}$$

From Eq. (3.23) the PF is found as

$$\text{PF} = \frac{I_{s1}}{I_s}\cos(-\alpha) = \frac{2\sqrt{2}}{\pi}\cos\alpha \tag{10.7}$$

b. $\alpha = \pi/3$

$$V_{\text{dc}} = \frac{2V_m}{\pi}\cos\alpha = 54.02\ \text{V} \qquad \text{and} \qquad V_n = 0.5\ \text{pu}$$

$$V_{\text{rms}} = \frac{V_m}{\sqrt{2}} = V_s = 120\ \text{V}$$

$$I_{s1} = \left(2\sqrt{2}\,\frac{I_a}{\pi}\right) = 0.90032I_a \qquad \text{and} \qquad I_s = I_a$$

$$\text{HF} = \left[\left(\frac{I_s}{I_{s1}}\right)^2 - 1\right]^{1/2} = 0.4834 \quad \text{or} \quad 48.34\%$$

$$\phi_1 = -\alpha \qquad \text{and} \qquad \text{DF} = \cos(-\alpha) = \cos\frac{-\pi}{3} = 0.5$$

$$\text{PF} = \frac{I_{s1}}{I_s}\cos(-\alpha) = 0.45\ (\text{lagging})$$

Note: The fundamental component of input current is always 90.03% of I_a and the HF remains constant at 48.34%.